

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	Specification for Fire & Gas Detection System- BBOU/ELEP/AKAB
Document Number	EA-BFPCL-GPMC-GGPL-FEED-ICS-DSH-022
Document Revision	R02
Document Status	Issued for Review
Originator	Paul Roy
Security Classification	Restricted
Issue Date	19-Oct-22

Specification for Fire & Gas Detection System- BBOU/ELEP/AKAB

R02	19.10.22	Re-Issued for Review	P. Roy	P. Olukah	A. Duru	
R01	05.07.22	Issued for Review	P. Roy	P. Olukah	W. Irem	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R01	05.07.22	Issued for Review
R02	19.10.22	Re-issued for Review after incorporating TCE comments

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

Restricted

This document is made available subject to the condition that the recipient will neither use nor disclose the contents except as agreed in writing with the copyright owner. Copyright is vested in Brass Fertilizer and Petrochemical Company Limited (BFPCL) © All rights reserved.

Neither the whole nor any part of this document may be reproduced or distributed in any form or by any means (electronic, mechanical, reprographic, recording, or otherwise) without the prior written consent of the copyright owner.

SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-ICS-DSH-022

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMG-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)

1. INTRODUCTION	6
2. DEFINITIONS	8
3. GENERAL	9
4. TYPE OF DETECTORS	10
5. POINT TYPE IR DETECTORS	11
6. OPEN PATH IR DETECTORS	13
7. FLAME DETECTOR	15
8. MANUAL ALARM CALL POINT	16
9. SMOKE AND HEAT DETECTORS	17
10. HOOTER AND BEACON	18
11. TAG LIST FOR DETECTORS,MAC AND HOOTER/BEACON	19
<i>APPENDIX - A</i>	20
<i>APPENDIX – B</i>	21

Abbreviations

Abbreviation	Meaning
ASME	American Society of Mechanical Engineers
API	American Petroleum Institute
IEC	International Electro technical Commission
ISA	Instrument Society of America
ITU	International telecommunication Union
SCADA	Supervisory Control and Data Acquisition System
MAC	Manual Alarm Call Point

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opubene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

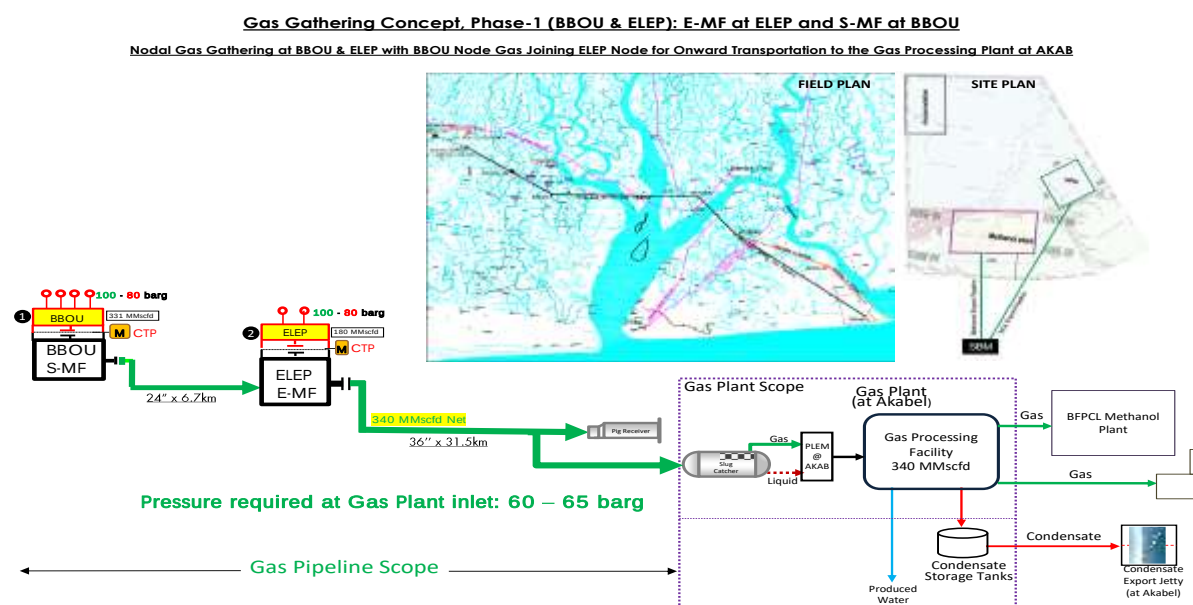


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

This specification covers the minimum requirements for the design, manufacture, assembly, inspection, testing, certification and delivery of F & G Instruments complete with accessories for BBOU, ELEP and AKAB pipeline.

2. Definitions

LEL	Low Explosive Limit: The lowest concentration (percentage) of Hydrocarbon vapor in air capable of producing a flash of fire in presence of an ignition source (arc, flame, heat).
Autolgnition temperature	Lowest temperature of the substance at which it will spontaneously ignite in a normal atmosphere without an external source of ignition Specifications/Datasheets Hydrocarbon Detection System for Petroleum Oil Installations 4
Flammable Gas	A flammable gas that burns when comes in to contact with heat or flame.
Flash point	Flash point of a volatile liquid is the lowest temperature at which it can vaporize to form an ignitable mixture in air
ppm	Part per million (ppm). It denotes one part per 1,000,000 parts, one part in 10^6 , or a value of 1×10^{-6} .

3. General

- 3.1. All gas detectors shall follow Shell DEP standards.
- 3.2. Fire and Gas Detectors shall have a minimum ingress protection of IP65 in accordance with IEC 60529. All instruments shall be intrinsically safe or explosion proof and shall be certified to specific hazardous area class by statutory body such as ATEX, PTB.

3.3. Materials of Construction

Materials of construction for all Fire and Gas Detectors shall be Stainless Steel. Material of Construction of Manual Alarm Call Point and Hooter / Beacon shall be Marine Graded Aluminium, if Stainless Steel is not available.

The following materials shall not be used in contact with the product:

3.4. Hazardous Area Requirement

- 3.4.1. Classification of Hazardous Areas shall be Zone1, IIA/IIB, T4.

All instrumentation located or partially located in the hazardous area shall be suitable for classified hazardous area. All instruments and Junction Boxes shall be certified as Increased Safety (Exe). Power supply shall be switched off before opening the Instrument or Junction Box. ATEX/PTB approved certification shall be supplied for all electrical / electronic equipment in the hazardous area.

- 3.5. All hydrocarbon detectors shall be SIL-2 certificated from TUV/Exida.
- 3.6. Accessibility: All instruments shall be accessible from grade or platform for measurement. In case of gas service where instruments are mounted above the taps, the installation of platform shall be considered, if accessibility is not available.

4. Type of Detectors

- 4.1. IR based integrated (sensor cum transmitter) point type Hydrocarbon detectors for measurement, signal transmission and monitoring of LEL levels in application area.
- 4.2. IR based Open Path detectors for measurement, signal transmission and monitoring of LEL levels in application area.
- 4.3. Manual Alarm Call Points shall be installed at strategic locations.
- 4.4. The detector location shall be as per fire and gas system location plan.
- 4.5. All detectors shall be connected to the safety system for the reception, indication, control and relay signals originating from various fire detector or call points connected to it, for activation of fire alarm sounders and Visual alarms as per cause and effect chart. The F&G system shall be designed for early and reliable detection of gas leakages, which otherwise could lead accumulations of gas, so shutdown actions can be initiated before flammable concentrations reach ignition sources. Likewise if gasses already have been ignited fires are to be detected sufficiently early so shutdown actions can be initiated to minimize risk to personnel and the plant.
- 4.6. The hydrocarbon detectors along with hooter/beacon and Manual Alarm Call point shall be integrated with SIL 3 certified SCADA system.
- 4.7. Hydrocarbon detection system is required for continuous monitoring of combustible Hydrocarbon where hydrocarbon product leaks can happen in the platform. The key component of detection shall be sweet gas Methane (CH₄).
- 4.8. Audio-Visual Alarm to be generated in the SCADA system to alert the Operator & the hooter cum beacon should operate in the field to alert the field operator as soon as the % LEL / ppm level as sensed by the detector in the field crosses the set level.
- 4.9. The hooter / beacon in the field will be activated in the field and shall be reset from the remote location
- 4.10. Alarms shall be of minimum three levels High, High – High & Fault and the controllers shall have provision for Inhibiting the HC detector outputs during maintenance and calibration. System shall have clear distinguishing between process and fault alarms.
- 4.11. High alarm set point for point type of gas detectors shall be set at 20% LEL and the High-High alarm to be set at 60% LEL (adjustable)
- 4.12. High alarm set point for open path gas detectors shall be set at 0.1 LEL. and the High-High alarm to be set at 0.5 LEL.m (adjustable)
- 4.13. All cable glands shall be of SS 316.

5. Point Type IR Detectors

Sr. No.	Parameters	Minimum Requirement
1	Sensor type	Infrared absorption sensor technology based universal type explosion-proof Hydrocarbon sensor with transmitter. (Sensor should be smart type & should memorize calibration curves for the transmitter to recognize). Sensor and Transmitter shall be SIL-2 certified.
2	Principle of operation	Multi-beam, dual compensated, non-focusing infrared absorption. Temperature compensation shall be in-built.
3	Function	Shall be able to detect Hydrocarbon in the range of 0-100% LEL
4	Hydrocarbon Detection	Configurable library having Methane, propane, propylene, Ethane, Butane, Hexane, Pentane & Benzene/R-LNG, covering Hydrocarbons vapours (from C1-C12)
7	Construction	Flameproof, SS 316 body with dust/weather protection for outdoor installation. No external terminal box shall be provided for further cabling (flying leads are not acceptable).
8	Optical performance	Correct operation up to 75% obscuration, the same shall be configurable with facility for dirty optics warning. Must have provision for heated optics to avoid condensation.
9	Input Power	24V DC nominal (18-30V DC)
10	Output	3-wire system, Linear 4-20mA DC (isolated / non-isolated) rated at 600 ohms loop resistance at 24V DC + HART. Configurable detector fault signal (0 mA), beam blockage, maintenance signal at 23.2 mA over range etc. Potential free contact output for LEL alarm.
11	Overall accuracy	$\pm 2\%$ full-scale ($\leq 50\%$ LEL) $\pm 5\%$ full-scale ($> 50\%$ LEL)
12	Repeatability	+/- 2% Full Scale Deflection (FSD)
13	Zero drift	2 % FSD per year maximum.

Sr. No.	Parameters	Minimum Requirement
14	Response time	90% of Hydrocarbon Vapour reading less than 2 Seconds.
15	Warm-up Time	Maximum 2 minutes from cold start-up.
16	Visual Status display	LCD/LED/OLED indication for Local indication parameters e.g.configuration,displayofconcentrationofthe Hydrocarbon Vapour detected (% LEL) etc. While LED indication for– Normal, Fault condition & Hydrocarbon Vapour detected condition
17	Diagnostics/Self Check	Continuous self-check for immediate detection of internal failures. Following Minimum Diagnostics shall be available in Host System/SCADA Sensor Power fault, IR Source fault, Sensor life expired, Zero Cal fault/ Span Cal fault, IR Low, IR detector fault, Electronics fault, Sensor missing, Sensor overrange etc.
18	RFI /EMI protection compliance	EN50270:1999 Type 2 (EMC- electrical apparatus for the detection of combustible gas) /latest revision of EMC directive 89/336/EEC.

6. Open Path IR Detectors

Sr. No.	Parameters	Minimum Requirement
1	Sensor type	Line of Sight Infra red absorption technique (xenon flash lamp) having field replaceable transmitter and receiver units. Field Alignment: By local digital display and adjustable mounting arms. Sensor and Transmitter shall be SIL-2 certified.
2	Transmitter / Receiver Assembly	Microprocessor based SMART type
3	Range	0- 5 LEL.m
4	Hydrocarbon Detection	Configurable library having Methane, propane, propylene, Ethane, Butane, Hexane, Pentane & Benzene /R-LNG, covering Hydrocarbons vapours (from C1-C12)
5	Construction	Flameproof, SS 316 body with dust/weather protection for outdoor installation. No external terminal box shall be provided for further cabling (flying leads are not acceptable).
6	Optical performance	Correct operation up to 75% obscuration, the same shall be configurable with facility for dirty optics warning. Must have provision for heated optics to avoid condensation.
7	Input Power	24V DC nominal (18-30V DC)
8	Output	Linear 4-20mA DC (isolated / non-isolated) rated at 600 ohms loop resistance at 24V DC + HART Configurable detector start up / fault signal (0 mA), Test gas / setup (1.5mA), beam blockage (2.0 mA), maintenance signal at 21.7mA over range etc. Potential free contact output for LEL-m alarm.
9	Overall accuracy	$\leq \pm 5\%$ of full scale for each scale or $\pm 10\%$ of applied gas, whichever is greater hydrocarbon vapour concentration
10	Repeatability	+/- 5% Full Scale Deflection(FSD)
11	Zero drift	2 % FSD per year maximum.

Sr. No.	Parameters	Minimum Requirement
12	Response time	90% of Hydrocarbon Vapour reading less than 2 Seconds.
13	Warm-up Time	Maximum 2 minutes from cold start-up.
14	Visual Status display	LCD/LED/OLED indication for Local indication parameters.
15	Diagnostics/Self Check	Continuous self-check for immediate detection of internal failures. Following Minimum Diagnostics shall be available in Host System/SCADA Sensor Power fault, IR Source fault, Sensor life expired, Zero Cal fault/ Span Cal fault, IR Low, IR detector fault, Electronics fault, Sensor missing, Sensor overrange etc.
16	RFI /EMI protection compliance	EN50270:1999 Type 2 (EMC- electrical apparatus for the detection of combustible gas) /latest revision of EMC directive 89/336/EEC.

7. Flame Detector

Type (U.V./I.R)	UV/IR Flame Detector
Distance	80-100 Ft.
Spectral Sensitivity	UV – 0.185 to 0.245 microns and IR – 4.2 to 4.7 microns
Cone Of Vision	Minimum 120 Deg.
Flame Sensitivity	0.09m ² gasoline fire at a distance between 15 to 19 mtrs.
Response Time	Field selectable between 800ms to 1sec.
Signal Output	4 – 20mA DC
Optical Integrity (Oi Test)	Yes
Reverse Polarity Protection	Yes
Detector Enclosure Material	Copper Free Aluminium
Electrical Connection	Terminals For 1,5 mm ² Wire
Cable Entry	(M 20 x 1.5 NPTF)
Enclosure Protection	Min: IP: 65
Electrical Protection	Suitable for use in Zone 1 Gr. IIA & IIB, T3
Transmission Supply	24 V DC
Power Consumption	< 2 W Nominal, 5 W Max.
Accessories	Surface mounting Swivel type Assembly
Electrical Certification	CENELEC / ATEX / BASEEFA

8. Manual Alarm Call Point

Sr. No.	Parameters	Minimum Requirement
1	Type of Manual Alarm Point	Push / Break Glass fitted with end of line resistors suitable for line monitoring.
2	Enclosure material	Anti-static UV stable Glass Reinforced Polyester/ SS 316 / Aluminium
3	Switch Type / Rating	1 pole Changeover / 2 Amps, 24VDC resistive.
4	Finish	Red Epoxy Painted.(DIN-RAL 3000 or equal)
5	Label	All call points shall be supplied with a label 'Fire Alarm" in English Language.
6	Certification	Explosion Proof Exd, IP-65, Zone1, IIA/IIB, T4

9. Smoke and Heat Detectors

- 9.1.** Multi sensor shall have combined & individual features of smoke detector and thermal sensor (heat detector). Smoke activity shall be monitored by photoelectric sensing technology in a single detector / base assembly. Heat detection shall be fixed with selectable rate of rise settings. Detectors shall be analogue type and shall be connected to SCADA/PLC system.
- 9.2.** The detector shall be able to sense incipient fire by detecting the presence of visible and invisible products of combustion like wood, paper, ammonia processing paper, cloth, PVC, bakelite, nylon, foam, acrylic, thermocol, Photo film, nylon, polyester, painted sheets, Teflon, leather etc.
- 9.3.** The sensitivity of the detector shall not vary with change in ambient temperature, humidity, pressure or permissible voltage variation. Its performance shall not be affected by an air current of 5 m/sec. It shall have an inbuilt arrangement such that puffs of smoke or hot air pockets do not inadvertently trigger the alarm. The detector shall be protected against dust accumulation / ingress IP-42. It shall have insect resistant screen to prevent nuisance alarms.

10. Hooter and Beacon

Sr. No.	Parameters	Minimum Requirement
1	Type of Hooter and Beacon	Combined Hooter / Beacon Assembly shall be provided.
2	Enclosure material	Anti-static UV stable Glass Reinforced Polyester/ SS 316 / Aluminium
3	Hooter/Beacon Intensity	Hooter shall have a decibel rating 10dBA above background noise level with a maximum 105 dBA, 10 feet away. Visual warning device shall be a high visibility flashing or rotating type lights. The beacon shall be of stroboscopic type and shall be of sufficient intensity to provide visibility and clear contrast during full daylight.
4	SCADA interface	SCADA shall provide the following for each Beacon & Hooter assembly: i) One Wet contact for high and One Potential free contact for high-high from SCADA for activating Hooter / Beacon.
5	Accessories	All accessories required for installation of hooter and Beacon shall be supplied by vendor. Vendor shall furnish installation details of Hooter and beacon after placement of order.
6	Power Supply and Signal Rating	Both Beacon and hooter units shall operate at 240VAC as specified in the data sheet and shall be certified explosion proof (flame proof) when installed in hazardous area. B19. Hooter/ Beacon shall be flameproof and shall be supplied with Acknowledge / reset push-buttons which shall be rated for 0.5 A/ 24V DC, 2 A as per the requirement specified in the data sheets.
7	Label	Hooter / Beacon shall be supplied with a label 'Fire Alarm" in English Language.
8	Certification	Explosion Proof Exd, IP-65, Zone1, IIA/IIB, T4

11. Tag list for Detectors, MAC and Hooter / Beacon

Sr. No.	Tag No.	Type of Detectors/ Instruments
1	BBOU-AE/AT-1001	Point Type Gas Detector
2	BBOU-AE/AT-1002	Point Type Gas Detector
3	BBOU-AE/AT-1003T	Open Path Gas Detector (Transmitter)
4	BBOU-AE/AT-1003R	Open Path Gas Detector (Receiver)
5	BBOU-AE/AT-1004T	Open Path Gas Detector (Transmitter)
6	BBOU-AE/AT-1004R	Open Path Gas Detector (Receiver)
7	BBOU-MAC-1001A	Manual Alarm Call Point (Field ESD)
8	BBOU-MAC-1002A	Manual Alarm Call Point (Field ESD)
9	BBOU-MAC-1003A	Manual Alarm Call Point (Field ESD)
10	BBOU-MAC-1004A	Manual Alarm Call Point (Field ESD)
11	BBOU-HB-1004	Hooter and beacon combine
12	BBOU-FD-1001	UV/IR Flame Detectors
13	BBOU-FD-1002	UV/IR Flame Detectors
14	ELEP-AE/AT-1001	Point Type Gas Detector
15	ELEP-AE/AT-1002	Point Type Gas Detector
16	ELEP-AE/AT-1003T	Open Path Gas Detector (Transmitter)
17	ELEP-AE/AT-1003R	Open Path Gas Detector (Receiver)
18	ELEP-AE/AT-1004T	Open Path Gas Detector (Transmitter)
19	ELEP-AE/AT-1004R	Open Path Gas Detector (Receiver)
20	ELEP-MAC-1001A	Manual Alarm Call Point (Field ESD)
21	ELEP-MAC-1002A	Manual Alarm Call Point (Field ESD)
22	ELEP-MAC-1003A	Manual Alarm Call Point (Field ESD)
23	ELEP-MAC-1004A	Manual Alarm Call Point (Field ESD)
24	ELEP-HB-1004	Hooter and beacon combine
25	ELEP-AE/AT-1005	Point Type Gas Detector
26	ELEP-AE/AT-1006T	Open Path Gas Detector (Transmitter)
27	ELEP-AE/AT-1006R	Open Path Gas Detector (Receiver)
28	ELEP-MAC-1005A	Manual Alarm Call Point (Field ESD)
29	ELEP-FD-1001	UV/IR Flame Detectors
30	ELEP-FD-1002	UV/IR Flame Detectors
31	AKAB-AE/AT-1001	Point Type Gas Detector
32	AKAB-AE/AT-1002T	Open Path Gas Detector (Transmitter)
33	AKAB-AE/AT-1002R	Open Path Gas Detector (Receiver)
34	AKAB-MAC-1001A	Manual Alarm Call Point (Field ESD)
35	AKAB-HB-1003	Hooter and beacon combine
36	AKAB-FD-1001	UV/IR Flame Detectors
37	AKAB-FD-1002	UV/IR Flame Detectors
38	RTU-SHD-1001	Smoke and Heat Detectors
39	RTU-SHD-1002	Smoke and Heat Detectors

APPENDIX - A

ENVIRONMENTAL CONDITIONS

Environmental Data

Environmental Conditions

Air Temperature : 18°C to 38°C.

Relative Humidity

Parameters	Values
Mean	81%
Minimum	51%
Maximum	95%

Seismic Zone:

The project area is classified as seismic zone 0

APPENDIX – B

CODES and STANDARDS

Applicable standards referred below shall be of the latest editions:

Shell DEP standards

IEC-60079	Electrical apparatus for Explosive Gas Mixture.
IEC-60529	Classification of degree of protection provided by enclosures.
IEC-801	Electromagnetic compatibility for Industrial process measurement & control equipment
IEC-61508	Functional Safety of electrical/electronic/ programmable electronic safety related system.
IEC-61511	Functional safety – Safety instrumented system for the Process Industry.
BS EN-50054	Electrical apparatus for the detection & measurement of combustible gases – General requirements & Test methods.
BS EN-50057	Performance requirements for Group II apparatus indicating up to 100% LEL.
EN 50270	Electrical apparatus for the detection and measurement of combustible gases, toxic gases and oxygen.
EN 50241	Specification for open-path apparatus for the detection of gases and vapors
DIN-50049	Document on material testing.
MIL-HDBK 217	Reliability prediction for Electronic Equipment